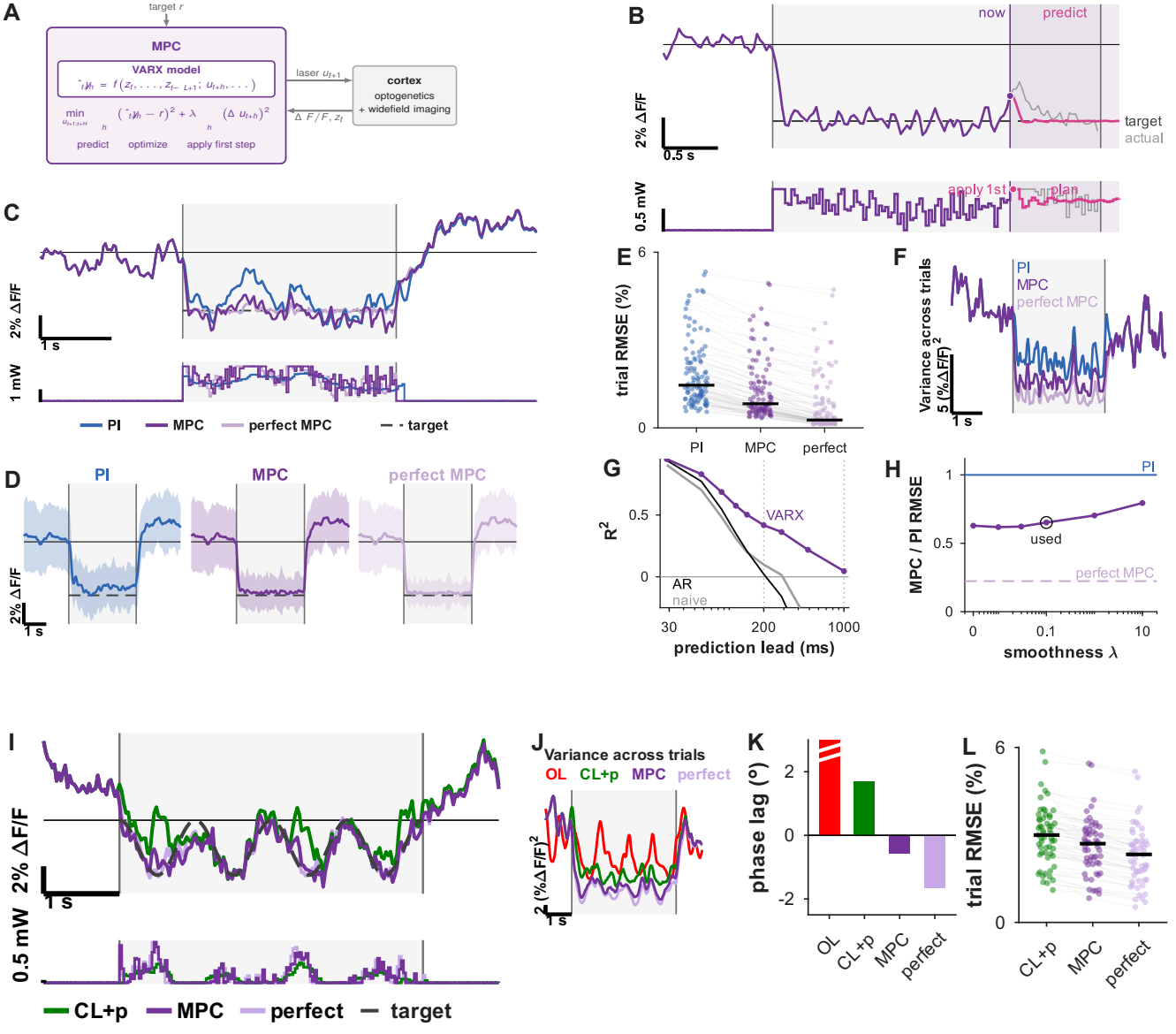

A
target r
MPC
VARX model
predict
optimize
apply first step
laser u_{t+1}
cortex
optogenetics
+ widefield imaging
$\Delta F/F, z_t$
B
now
predict
target
actual
2% ΔF/F
0.5 s
0.5 mW
apply 1st
plan
C
2% ΔF/F
1 s
1 mW
PI
MPC
perfect MPC
target
D
PI
MPC
perfect MPC
2% ΔF/F
1 s
E
trial RMSE (%)
6
0
PI
MPC
perfect
F
Variance across trials
PI
MPC
perfect MPC
5 (%ΔF/F)^2
1 s
G
R^2
0.5
0
VARX
AR
naive
30
200
1000
prediction lead (ms)
H
MPC / PI RMSE
1
0.5
0
PI
used
perfect MPC
0
0.1
10
smoothness λ
I
2% ΔF/F
1 s
0.5 mW
CL+p
MPC
perfect
target
J
Variance across trials
OL
CL+p
MPC
perfect
2 (%ΔF/F)^2
1 s
K
phase lag (°)
2
0
-2
OL
CL+p
MPC
perfect
L
trial RMSE (%)
6
0
CL+p
MPC
perfect